

Progress Test

AERO40006 - Mathematics 1

80/100

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Answer All Questions

Question 1

Part (a)

You are a rice producer and your production capacity for today is 1,200 kg which you can sell for 0.40 \$ per kg. You can decide to wait one week or more to produce and to sell in order to increase your revenues. Your production capacity increases by 100 kg per week however the selling price per kg will go down every week by 0.02 \$. When do you need to produce and to sell in order to maximise your revenues?

15 [15%]

$$\begin{aligned} (a) \quad R(w) &= (0.4 - 0.02w)(1200 + 100w) \\ &= -2w^2 + 16w + 480 \end{aligned}$$

$$\therefore R'(w) = -4w + 16$$

$$\therefore \Delta R(w) = 16^2 - 4(-2) \times 480 = 4096$$

$$R''(w) = -4 < 0$$

$$\therefore \text{When } -4w + 16 = 0, \quad R(w) \text{ maximum}$$

$$\therefore \text{When } w = 4 \text{ (weeks),}$$

$$R(w)_{\max} = 512 (\$)$$

Part (b)

The critical insulation thickness of a tube is defined as the value of t which gives minimum resistance R where

$$R = \frac{\ln(t/t_i)}{2\pi k} + \frac{1}{2\pi t h}.$$

$k > 0$ is the thermal conductivity, h is the heat transfer coefficient and t_i is the inner thickness of insulation tube. Show that the critical thickness is

15 [15%]

$$t = \frac{k}{h}.$$

$$(b) \quad R = \frac{\ln(\frac{t}{t_i})}{2\pi k} + \frac{1}{2\pi h} \frac{1}{t} = \frac{\ln(t)}{2\pi k} - \frac{\ln(t_i)}{2\pi k} + \frac{1}{2\pi h} \frac{1}{t}$$

$$\therefore R' = \frac{1}{2\pi k} \frac{1}{t} - \frac{1}{2\pi h} \frac{1}{t^2}$$

When R minimum,

$$R' = \frac{1}{2\pi} \left(\frac{1}{k} \frac{1}{t} - \frac{1}{h} \frac{1}{t^2} \right) = 0$$

$$\therefore \frac{1}{k} - \frac{1}{h} \frac{1}{t} = 0$$

$$\therefore \frac{1}{h} = \frac{1}{k} t$$

$$\therefore t = \frac{k}{h}$$

Question 2

Part (a)

Evaluate the integral $I = \int_0^{\pi} \frac{x \sin(x)}{1 + \cos^2(x)} dx$.

Q [10%]

$$(a) \quad I = \int_0^{\pi} \frac{x \sin(x)}{1 + \cos^2(x)} dx$$

$$\text{Let } u = \pi - x$$

$$\therefore \frac{du}{dx} = -1 \quad \therefore dx = -du$$

$$\therefore I = - \int_{\pi}^0 \frac{(\pi - u) \sin(u)}{1 + \cos^2(u)} du = \int_0^{\pi} \frac{(\pi - u) \sin(u)}{1 + \cos^2(u)} du$$

$$= \pi \int_0^{\pi} \frac{\sin(u)}{1 + \cos^2(u)} du - I$$

$$\therefore I = \frac{\pi}{2} \int_0^{\pi} \frac{\sin(u)}{1 + \cos^2(u)} du$$

$$\text{Let } t = \cos(u)$$

.....

Part (b)

Let us consider the complex number $z = -\frac{\sqrt{2}}{1+i}e^{i\frac{\pi}{3}}$. Are the following statements true or false? Briefly justify your answers.

1. $|z| = 1$. 10 [2.5%]
2. $z = -(1-i)e^{i\frac{\pi}{3}}$. [2.5%]
3. $-\frac{\pi}{2}$ is an argument of z . [2.5%]
4. $z = e^{i\frac{13\pi}{12}}$. [2.5%]

$$\begin{aligned}
 1. \quad z &= -\frac{\sqrt{2}}{1+i} e^{i\frac{\pi}{3}} = -\frac{\sqrt{2}}{1+i} \left[\cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right) \right] \\
 &= -\frac{\sqrt{2}}{1+i} \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) \\
 &= -\frac{\sqrt{2}}{2} \left(\frac{1+\sqrt{3}i}{1+i} \right) = -\frac{\sqrt{2}}{2} \left(\frac{(1+\sqrt{3}i)(1-i)}{(1+i)(1-i)} \right) \\
 &= -\frac{\sqrt{2}}{2} \left(\frac{1+\sqrt{3}-i+\sqrt{3}}{2} \right) = -\frac{\sqrt{2}}{4} [(1+\sqrt{3}) + (-1+\sqrt{3})i] \\
 &= -\left(\frac{\sqrt{6}+\sqrt{2}}{4} \right) - \left(\frac{\sqrt{6}-\sqrt{2}}{4} \right)i
 \end{aligned}$$

$$\therefore |z| = \left(\frac{\sqrt{6}+\sqrt{2}}{4} \right)^2 + \left(\frac{\sqrt{6}-\sqrt{2}}{4} \right)^2 = 1^2 = 1 \quad \therefore |z| = 1 \text{ is true}$$

$$\begin{aligned}
 2. \quad z &= -\frac{\sqrt{2}}{1+i} e^{i\frac{\pi}{3}} = -\frac{\sqrt{2}(1-i)}{(1+i)(1-i)} e^{i\frac{\pi}{3}} = -\frac{\sqrt{2}}{2}(1-i)e^{i\frac{\pi}{3}} \neq -(1-i)e^{i\frac{\pi}{3}} \\
 &\therefore z = -(1-i)e^{i\frac{\pi}{3}} \text{ is false}
 \end{aligned}$$

$$3. \quad \because z = |z|e^{i\theta} \quad \therefore e^{i\theta} = -\left(\frac{\sqrt{6}+\sqrt{2}}{4} \right) - i\left(\frac{\sqrt{6}-\sqrt{2}}{4} \right) = \cos\theta + i\sin\theta$$

$$\therefore \cos\theta = -\left(\frac{\sqrt{6}+\sqrt{2}}{4} \right), \quad \sin\theta = -\left(\frac{\sqrt{6}-\sqrt{2}}{4} \right)$$

Part (b)

Let us consider the complex number $z = -\frac{\sqrt{2}}{1+i}e^{\frac{i\pi}{3}}$. Are the following statements true or false? Briefly justify your answers.

1. $|z| = 1$. [2.5%]

2. $z = -(1-i)e^{\frac{i\pi}{3}}$. [2.5%]

3. $-\frac{\pi}{2}$ is an argument of z . [2.5%]

4. $z = e^{\frac{13i\pi}{12}}$. [2.5%]

$$\therefore \arccos\left(\frac{-\sqrt{6}-\sqrt{2}}{4}\right) = \frac{11}{12}\pi \text{ or } \frac{13}{12}\pi \text{ or } \dots$$

$$\arcsin\left(\frac{-\sqrt{6}+\sqrt{2}}{4}\right) = -\frac{1}{12}\pi \text{ or } \frac{13}{12}\pi \text{ or } \dots$$

$$\therefore \arg(z) = \theta = \frac{13}{12}\pi \quad (0 \leq \theta < 2\pi)$$

$$\therefore -\frac{\pi}{2} \text{ is not an argument of } z$$

4. $\therefore \theta = \frac{13}{12}\pi$

$$\therefore z = |z|e^{i\theta} = e^{\frac{13}{12}i\pi} \text{ is true}$$

Part (c)

Evaluate $(1+i)^{1000}$.

10 [10%]

$$\begin{aligned}(1+i)^{1000} &= \sum_{r=0}^{1000} {}^{1000}C_r 1^{1000-r} (i)^r \\ &= \sum_{r=0}^{1000} {}^{1000}C_r (i)^r\end{aligned}$$

$$(1+i)^2 = 1 - 1 + 2i = 2i$$

$$(1+i)^3 = 2i - 2$$

$$(1+i)^4 = -4$$

$$(1+i)^5 = -4 - 4i$$

$$(1+i)^6 = (-4 - 4i)(1+i) = -4 - 4i - 4i + 4 = -8i$$

$$(1+i)^7 = -8i + 8$$

$$(1+i)^8 = (-8i + 8)(1+i) = -8i + 8 + 8 + 2i = 16$$

...

$$\therefore \text{if } n \text{ is even and } n > 0, (1+i)^n = 2^{\frac{n}{2}} (i)^{\frac{n}{2}}$$

$$\text{if } n \text{ is odd and } n > 0, (1+i)^n = (1+i) \left[2^{\frac{n-1}{2}} (i)^{\frac{n-1}{2}} \right]$$

 $\therefore 1000$ is even

$$\begin{aligned}\therefore (1+i)^{1000} &= 2^{\frac{1000}{2}} (i)^{\frac{1000}{2}} = 2^{500} i^{500} = 2^{500} (i^2)^{250} \\ &= 2^{500} (-1)^{250} = 2^{500}\end{aligned}$$

Part (d)

Find the 3rd-order Taylor polynomial of $f(x, y) = e^{x^2+y}$ around $(x, y) = (0, 0)$.

0 [10%]

$$f(x, y) = e^{x^2+y} = e^{x^2} e^y$$

$$f_x(x, y) = 2e^{x^2+y}$$

$$f_y(x, y) = e^{x^2+y}$$

$$\therefore f_{xy}(x, y) = f_{yx}(x, y) = 2e^{x^2+y}$$

$$\therefore f_{xx}(x, y) = 4e^{x^2+y}$$

$$f_{yy}(x, y) = e^{x^2+y}$$

$$\therefore f(x, y) = f(0, 0) + (x-0)f_x(0, 0) + (y-0)f_y(0, 0)$$

$$+ \frac{1}{2} [(x-0)^2 f_{xx}(0, 0) + 2(x-0)(y-0)f_{xy}(0, 0) + (y-0)^2 f_{yy}(0, 0)]$$

$$= 1 + 2x + y + \frac{1}{2} (4x^2 + 4xy + y^2)$$

$$= 1 + 2x + y + 2x^2 + \frac{1}{2}y^2 + 2xy$$

$$e^{x^2+y} = 1 + (x^2+y) + \frac{(x^2+y)^2}{2!} + \frac{(x^2+y)^3}{3!} + \dots$$

Question 3

A function $y(x)$ satisfies the equation

$$\frac{d^2y}{dx^2} + xy = 0$$

with the initial conditions $y(0) = 1$, $y'(0) = 0$.

Part (a)

Differentiate the equation n times to show that, for $n \geq 1$,

10 [10%]

$$\frac{d^{n+2}y}{dx^{n+2}} + n \frac{d^{n+1}y}{dx^{n+1}} = 0 \quad \text{at} \quad x = 0.$$

$$(a) \quad \frac{d^2y}{dx^2} = -xy$$

$$\text{At } x = 0$$

$$\therefore \frac{d^3y}{dx^3} = -y - x \frac{dy}{dx} = -1$$

$$\therefore \frac{d^4y}{dx^4} = -\frac{dy}{dx} - \frac{dy}{dx} - x \frac{d^2y}{dx^2} = -2 \frac{dy}{dx} - x \frac{d^2y}{dx^2} = -2 \frac{dy}{dx}$$

$$\therefore \frac{d^5y}{dx^5} = -2 \frac{d^2y}{dx^2} - \frac{d^2y}{dx^2} - x \frac{d^3y}{dx^3} = -3 \frac{d^2y}{dx^2} - x \frac{d^3y}{dx^3} = -3 \frac{d^2y}{dx^2}$$

.....

$$\therefore \frac{d^{n+2}y}{dx^{n+2}} = -n \frac{d^{n+1}y}{dx^{n+1}} - x \frac{d^{n+2}y}{dx^{n+2}}$$

$$\therefore \text{At } x = 0, \quad \frac{d^{n+2}y}{dx^{n+2}} + n \frac{d^{n+1}y}{dx^{n+1}} = 0$$

Part (b)

Express $y^{(3)}(0)$, $y^{(4)}(0)$, $y^{(5)}(0)$ and $y^{(6)}(0)$.

5 [5%]

$$\therefore y^{(3)}(0) = -y - x \frac{dy}{dx} = -1$$

$$y^{(4)}(0) = -2 \frac{dy}{dx} = 0$$

$$y^{(5)}(0) = -3 \frac{d^2y}{dx^2} = 3xy = 0$$

$$y^{(6)}(0) = -4 \frac{d^3y}{dx^3} = 4$$

Part (c)

Show that the Mclaurin series for $y(x)$ when $y(x)$ satisfies

$$\frac{d^2 y}{dx^2} + xy = 0$$

with $y(0) = 1$, $y'(0) = 0$ can be written as

$$y(x) = \sum_{i=0}^{\infty} c_i x^{3i}$$

where c_i are constants.

5 [5%]

$$y(x) = y(0) + y'(0)x + \frac{y''(0)}{2!}x^2 + \frac{y^{(3)}(0)}{3!}x^3 + \dots$$

$$= 1 + 0 + 0 - \frac{1}{3!}x^3 + 0 + 0 + \frac{4}{6!}x^6 + 0 + 0 - \frac{7}{9!}x^9 + \dots$$

$$= \sum_{i=0}^{\infty} \frac{y^{3i}(0)}{(3i)!} x^{3i}$$

Part (d)

Show that this series converges for all x .

10 [10%]

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{\left| \frac{y^{3n+1}(0)}{(3n+3)!} x^{3n+1} \right|}{\left| \frac{y^{3n}(0)}{(3n)!} x^{3n} \right|} &= \lim_{n \rightarrow \infty} \left| \frac{1}{3n+3} \frac{y^{3n+1}(0)}{y^{3n}(0)} x \right| \\
 &= \lim_{n \rightarrow \infty} \left| \frac{1}{3n+3} \frac{(-1)^{n+1} 3n-2+3}{(-1)^n 3n-2} x \right| \\
 &= \lim_{n \rightarrow \infty} \left| \frac{1}{3n+3} \frac{-3n+1}{3n-2} x \right| \\
 &\rightarrow 0 < 1
 \end{aligned}$$

$\therefore$ series converge for all x